Artificial Intelligence in Music Education: Trends, Challenges, and Future Directions

Chamil Arkhasa Nikko MAZLAN*
Music and Performing Arts Faculty
Universiti Pendidikan Sultan Idris
Tanjung Malim, Perak, Malaysia
jazzerlogy@gmail.com
https://orcid.org/0000-0002-0088-3849
*Corresponding author

Sajastanah Imam KONING
Music and Performing Arts Faculty
Universiti Pendidikan Sultan Idris
Tanjung Malim, Perak, Malaysia
https://orcid.org/0009-0007-7093-5863

Mohd Yusran Mohd YUSOFF
Music and Performing Arts Faculty
Universiti Pendidikan Sultan Idris
Tanjung Malim, Perak, Malaysia
yusran.yusoff@fmsp.upsi.edu.my

Dayang Siti Hazar Awang HASSIM
Colleg of Creative Arts
Universiti Teknologi MARA
Shah Alam, Selangor, Malaysia
dayangsitihazar@uitm.edu.my

Riyan HIDAYATULLAH
Fakultas Keguruan dan Ilmu Pendidikan
Universitas Lampung
Kota Bandar Lampung, Indonesia
https://orcid.org/0000-0001-7382-229X

Surasak JAMNONGSARN
Graduate Program in Thai and Asian Music, Faculty of Fine Arts
Srinakharinwirot University.
Bangkok, Thailand
https://orcid.org/0000-0002-1050-8639

***Abstract*— This study explores the integration of Artificial Intelligence (AI) in music education through a bibliometric analysis and scoping review. It identifies key research trends, influential works, and emerging themes related to AI-driven pedagogical applications. Beyond mapping scholarly output, the study synthesizes findings to highlight how AI enhances music learning through personalized feedback, operational efficiency, and improved music literacy. The analysis offers insights into AI's role in curriculum design and instructional strategies and proposes actionable implications for educators and researchers. The findings contribute to a deeper understanding of AI's impact and provide practical directions for advancing music education.**

Keywords*— *artificial intelligence, music education, bibliometric analysis, scoping review, ai pedagogy, dual systematic review

I. Introduction

The integration of Artificial Intelligence (AI) in music education has gained significant attention in recent years, driven by advancements in machine learning, natural language processing, and intelligent tutoring systems [1], [2]. AI technologies are transforming traditional pedagogical approaches by providing personalized learning experiences, automated assessment tools, and innovative composition techniques [3]. Researchers have explored AI's role in enhancing student engagement, optimizing instructional methodologies, and offering real-time feedback mechanisms in music education [4]–[8]. AI-driven platforms utilize deep learning models to analyze musical structures, assess students' performances, and generate adaptive learning content tailored to individual needs [9], [10]. Despite its potential, the adoption of AI in music education presents several challenges. The lack of AI literacy among educators, ethical concerns regarding AI-driven assessments, and the limitations of AI-generated music in capturing human expressiveness remain significant obstacles [11], [12]. Furthermore, issues related to data privacy, accessibility, and the cost of implementing AI-based systems require further examination [13], [14]. Addressing these challenges necessitates collaborative efforts between educators, AI developers, and policymakers to ensure that AI technologies effectively complement traditional music education practices. This study aims to provide a comprehensive analysis of AI applications in music education by examining existing literature and identifying key trends and research gaps. By synthesizing insights from recent scholarly contributions, this paper seeks to contribute to the ongoing discourse on AI's role in music education and propose future research directions. The subsequent sections will delve into a detailed discussion of AI applications, challenges, and prospects within music education, drawing on a range of interdisciplinary perspectives.

II. Methods

This study utilizes a Dual Systematic Review methodology, incorporating both bibliometric analysis and qualitative content analysis to examine trends, challenges, and future directions of Artificial Intelligence (AI) in music education. The bibliometric analysis evaluates publication patterns, source types, and keyword distributions, while the qualitative content analysis synthesizes thematic insights from the selected studies [15]–[22]. This approach ensures a structured and replicable method for identifying and synthesizing relevant scholarly contributions. Apart from that, this combination of method providing scoping review such as key themes and bibliometric add a quantitative nuances, trends and pattern [23]–[25]. Additionally, helping to expands future research [26], [27].

A. Data Collection and Search Strategy

The data for this study was collected from the Scopus database, a widely recognized bibliographic repository [28], [29]. The search strategy was designed to capture a broad spectrum of research in AI-driven music education using the following parameters, as illustrated in Fig. 1 below:

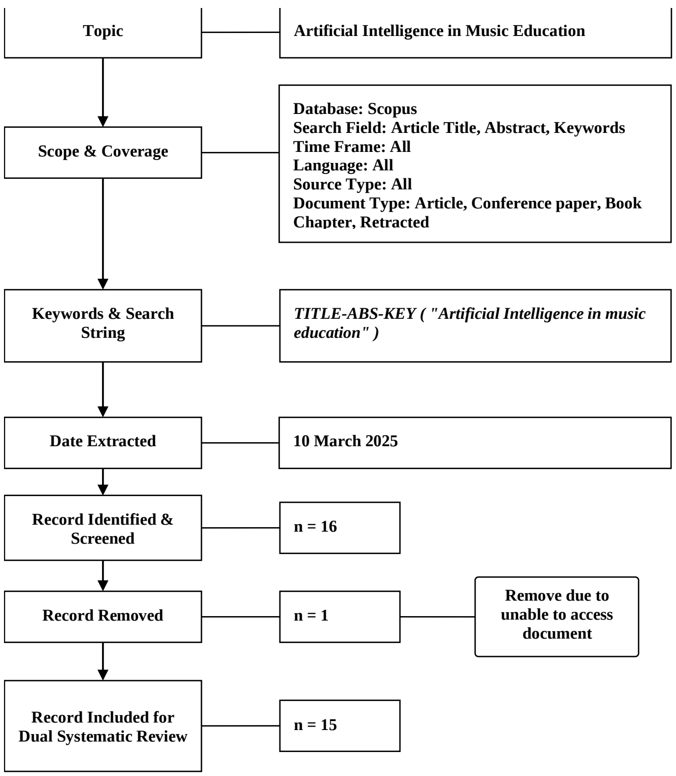

Fig. 1. Flow diagram of the search strategy.
Source: [19], [30]

The initial search yielded 16 records. However, one document was removed due to access limitations, resulting in 15 records being included for the systematic review. The search and selection process adhered to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, ensuring transparency and reproducibility [31]–[35].

B. Data Analysis and Tools

The bibliometric analysis was conducted using *BiblioMagika®* [36], a specialized software for bibliometric research. Key metrics analyzed included document profiling, publication trends, source title distribution, highly cited documents, and keyword analysis. These indicators provided insights into the research landscape and emerging topics in AI-driven music education. The scoping review was employed to synthesize qualitative insights from the selected studies. This process involved systematically categorizing findings related to AI applications, challenges, and pedagogical implications in music education. The integration of bibliometric analysis and scoping review ensured a structured methodological approach, capturing both quantitative publication trends and qualitative research insights [15], [17], [37], [38].

III. RESULTS

This section presents findings from the bibliometric analysis and scoping review of AI in music education, highlighting publication trends, citation metrics, document types, subject areas, and keyword distributions. Using BiblioMagika® [36], the analysis examines total publications, citation impact, and author contributions, offering insights into the progression of AI-related studies in music education. The systematic review explores the pedagogical implications, integration challenges, and future research opportunities of AI technologies in music education.

A. Document Profiles

Between 2013 and 2024, 15 publications on AI in music education were identified, with 258 citations and an h-index of 8. The average citation per paper was 17.20, and the citation rate was 23.45 per year. These publications involved 30 authors, averaging 2 authors per paper. Of the 15 papers, 13 received citations, averaging 19.85 citations per cited paper. The g-index and m-index were 15 and 0.615, respectively. Journal articles made up 54.86% of the 371 analyzed documents, followed by conference papers (31.89%) and book chapters (8.92%). Among the core 15 publications, nearly half were journal articles (46.67%), followed by conference papers (40.00%). Computer Science dominated 73.33% of the work, with Mathematics (40.00%), Engineering (26.67%), Social Sciences (20.00%), Arts and Humanities (13.33%), and smaller contributions from Physics and Astronomy and Psychology (each at 6.67%). This shows AI in music education is mainly driven by technical disciplines, with growing interest from educational and artistic fields

B. Publication Trends

Table I below shows the publication trends indicate a steady rise in research on AI in music education since 2019, peaking in 2022 and 2024, with 4 publications each year. The highest citation count (98 citations) occurred in 2022, reflecting a significant impact. The average citation per paper was 17.20, with 2023 showing the highest (35.50 citations per paper). The h-index (8) and g-index (15) suggest a growing scholarly influence. Despite fluctuations in yearly output, 2023 and 2024 demonstrate a sustained interest in the field.

TABLE I. PUBLICATION BY YEAR

Year	TP	NCA	NCP	TC	C/P	C/CP	*h*	*g*	*m*
2013	1	1	1	21	21.00	21.00	1	1	0.077
2019	2	2	2	19	9.50	9.50	2	2	0.286
2020	2	2	2	27	13.50	13.50	2	2	0.333
2022	4	11	4	98	24.50	24.50	4	4	1.000
2023	2	7	2	71	35.50	35.50	1	2	0.333
2024	4	7	2	22	5.50	11.00	2	4	1.000
Total	15	30	13	258	17.20	19.85	8	15	0.615

Notes: TP=total number of publications; NCA=Number of contributing authors; NCP=number of cited publications; TC=total citations; C/P=average citations per publication; C/CP=average citations per cited publication; h=h-index; g=g-index; m=m-index.
Source: Generated by the author(s) using biblioMagika® [36]

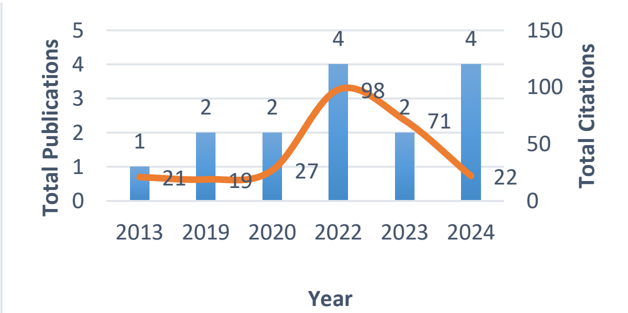

Fig. 2. Total publications and citations by year.
Source: Generated by the author(s) using biblioMagika® (Ahmi, 2024)

The bar chart in Fig. 2 above represents in the total number of publications per year, while the line graph illustrates total citations. The publication count peaked in 2022 and 2024 with four papers each, while 2022 had the highest citation count (98). A decline in citations is observed in 2024, indicating that recent publications may require time to accumulate citations.

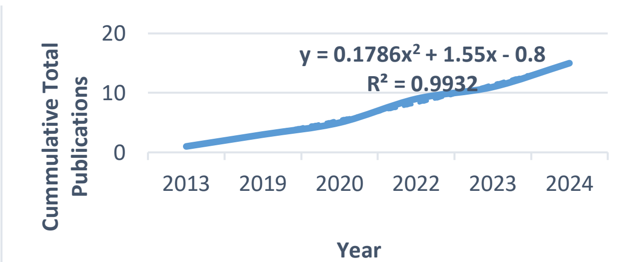

Fig. 3. Publication growth.
Source: Generated by the author(s) using biblioMagika® [36]

Fig. 3 above presents the cumulative growth of publications over time. The trend follows a quadratic pattern ($R^2 = 0.9932$), indicating a steady increase in research output on AI in music education. The growth rate accelerates after 2020, suggesting increasing academic interest in this field.

C. Highly Cited Documents

Among the literature on Artificial Intelligence in music education, several works have emerged as highly influential. The most cited study is by [1], titled Developments and Applications of Artificial Intelligence in Music Education, published in Technologies, which garnered 70 citations, averaging 23.33 citations per year. Close behind is the work of [2] on AI techniques in college music education, published in Computers and Electrical Engineering, with 63 citations and an annual citation rate of 15.75. Other significant contributions include [39], who proposed a decision-support system for evaluating AI and machine learning in networked music education environments (Soft Computing, 22 citations), and Holland (2013), whose critical review in Readings in Music and Artificial Intelligence has been cited 21 times over the years. [3] also made a notable impact with 19 citations for their article in the International Journal of Human-Computer Interaction, averaging 9.5 citations per year. Further down the list, [4] discussed AI applications in the field through Journal of Physics: Conference Series (16 citations), while [11] explored pedagogical uses in the Lecture Notes in Computer Science series (12 citations). Similarly, [5] presented findings in ICRIS 2020 proceedings (11 citations), and [12] examined AI integration supported by wireless networks in Mathematical Problems in Engineering (7 citations). Finally, [14] addressed the role of AI in supporting dyslexic learners, a study published in Communications in Computer and Information Science, also with 7 citations. Collectively, these top-cited documents reflect a diverse yet converging interest in how AI is transforming music education, ranging from pedagogical frameworks and technical systems to inclusive learning strategies.

D. Keyword Frequency Analysis

Fig. 4 below illustrates the top 15 author keywords used in the reviewed publications. The most frequently occurring keywords are "music education" (9 occurrences) and "artificial intelligence" (8 occurrences). Other keywords, such as "applications," "machine learning," and "chatbot," appear only once, indicating a more specialized focus in certain studies.

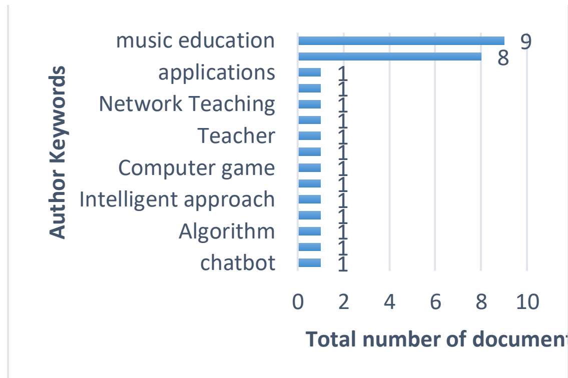

Fig. 4. Frequency distribution of top 15 author keywords.
Source: Generated by the author(s) using biblioMagika® [36]

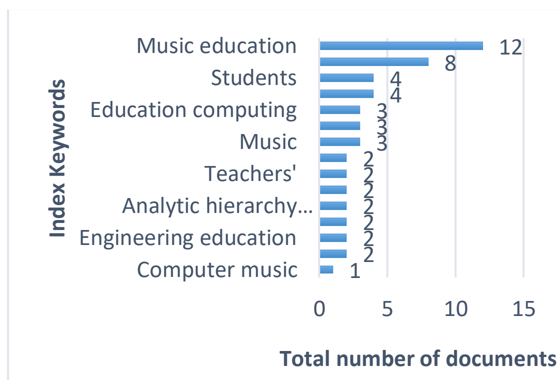

Fig. 5. Frequency distribution of top 15 index keywords.
Source: Generated by the author(s) using biblioMagika® [36]

Fig. 5 shows the top 15 index keywords in the reviewed publications. The most frequent terms are "music education" (12 occurrences) and "artificial intelligence" (8 occurrences). Other notable keywords include "students" and "artificial intelligence technologies" (4 occurrences each), highlighting key themes.

IV. Discussion

The integration of Artificial Intelligence (AI) in music education has opened new avenues for teaching, learning, and creative expression. This section discusses the key emerging trends in AI-driven music education, the challenges that educators and institutions face in implementing AI-based approaches, and potential future directions for research and practice. By analyzing recent scholarly contributions, this discussion aims to provide insights into how AI is shaping the field of music education and the implications for educators, students, and researchers.

A. Emerging Trends in AI-Driven Music Education

Recent scholarly contributions emphasize the transformative role of AI in music education. The integration of machine learning, deep learning, and intelligent tutoring systems has enabled adaptive and personalized learning experiences [1], [2]. AI-driven tools provide real-time feedback, allowing students to refine their skills efficiently. Studies have demonstrated the potential of AI in automated music composition, where intelligent systems assist in generating harmonies and melodies, fostering creativity among learners [4], [9]. Moreover, AI technologies are enhancing music theory instruction by analyzing complex musical structures and providing interactive learning experiences [39].

B. Challenges in Implementing AI in Music Education

Despite the advancements, significant challenges persist in AI-driven music education. One major issue is the lack of AI literacy among educators, which hinders effective implementation [5], [11]. Many educators require additional training to integrate AI-based tools into their teaching practices. Ethical concerns related to AI-generated music and assessments also arise, as questions about originality, authorship, and student evaluation persist [3]. Furthermore, AI-driven feedback mechanisms, while efficient, may lack the nuanced expressiveness that human instructors provide, raising concerns about the limitations of AI in music pedagogy [12].

C. Data Privacy and Accessibility Issues

AI-driven music education relies heavily on vast datasets for training and refining algorithms. However, data privacy remains a significant concern, particularly regarding the storage and usage of students' musical performances and personal information [14]. Accessibility is another pressing issue, as AI-based systems often require substantial financial investment, which may not be feasible for all institutions and learners. Disparities in technological infrastructure and access to AI-powered educational tools further exacerbate the digital divide in music education [2].

D. Practical Applications and the Impact of AI on Music Education

AI is transforming music education by making learning more personalized and responsive. Tools like ChatGPT and intelligent tutoring systems offer real-time feedback, helping students refine their skills and stay engaged. These adaptive technologies support customized lesson plans that meet diverse learner needs, making education more inclusive and effective [40], [41]. In terms of outcomes, AI-supported models such as CTCL have demonstrated measurable gains in students' music literacy, creativity, and cultural understanding. These models tap into AI's analytical power to deepen artistic expression and comprehension, highlighting its value as a meaningful pedagogical tool [42].

E. Future Directions

Future research should bridge the gap between AI capabilities and human musical expressiveness, focusing on developing AI systems that emulate interpretative and emotional aspects of music. Interdisciplinary collaboration among AI developers, music educators, and policymakers is essential for creating ethical guidelines and best practices. Enhancing AI literacy among educators through professional development programs can facilitate effective adoption of AI tools [1], [43]. Ensuring equitable access to AI-driven music education via open-source platforms and affordable technologies is crucial for democratizing AI's benefits in music learning.

V. CONCLUSIONS

The integration of AI in music education is transforming pedagogical approaches, enabling personalized learning, intelligent tutoring, and automated music composition. Despite these benefits, challenges like AI literacy, ethical concerns, and implementation costs must be addressed. This study highlights key trends, obstacles, and future directions in AI-driven music education by analyzing highly cited works. Collaboration among educators, researchers, and policymakers is crucial to ensure AI enhances, rather than replaces, traditional music education. Further research should explore AI's long-term effects on music pedagogy and its potential to improve accessibility worldwide.

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REFERENCES

[1] X. Yu, N. Ma, L. Zheng, L. Wang, and K. Wang, "Developments and Applications of Artificial Intelligence in Music Education," *Technol. 2023, Vol. 11, Page 42*, vol. 11, no. 2, p. 42, Mar. 2023, doi: 10.3390/TECHNOLOGIES11020042.

[2] J. Wei, K. Marimuthu, and A. Prathik, "College music education and teaching based on AI techniques," *Comput. Electr. Eng.*, vol. 100, p. 107851, May 2022, doi: 10.1016/J.COMPELECENG.2022.107851.

[3] P. ping Li and B. Wang, "Artificial Intelligence in Music Education," *Int. J. Human–Computer Interact.*, vol. 40, no. 16, pp. 4183–4192, 2024, doi: 10.1080/10447318.2023.2209984.

[4] S. Yuan, "Application and Study of Musical Artificial Intelligence in Music Education Field," *J. Phys. Conf. Ser.*, vol. 1533, no. 3, p. 032033, Apr. 2020, doi: 10.1088/1742-6596/1533/3/032033.

[5] F. Yang, "Artificial intelligence in music education," *Proc. - 2020 Int. Conf. Robot. Intell. Syst. ICRIS 2020*, pp. 483–484, Nov. 2020, doi: 10.1109/ICRIS52159.2020.00124.

[6] S. F. Arshad, M. A. Sulong, Z. Hashim, and C. A. N. Mazlan, "Perspective on Blended Learning for Instrumental Music Learning," *J. Posit. Sch. Psychol.*, vol. 6, no. 3, pp. 4936 – 4941–4936 – 4941, Mar. 2022, Accessed: Mar. 08, 2023. [Online]. Available: https://www.journalppw.com/index.php/jpsp/article/view/2700

[7] C. A. N. Mazlan *et al.*, "Musical Innovation: The Fusion of Chord Melody Style and Malay Traditional Music Lagu Melayu Asli," *Educ. Adm. Theory Pract.*, vol. 30, no. 4, pp. 2438–2448, Apr. 2024, doi: 10.53555/KUEY.V30I4.888.

[8] R. Hidayatullah, P. Tejapermana, C. Arkhasa, and N. Mazlan, "Music Performance Pedagogy: Self-Regulation in Contrabass Practice," *Virtuoso J. Pengkaj. dan Pencipta. Musik*, vol. 7, no. 2, pp. 184–198, Dec. 2024, doi: 10.26740/VT.V7N2.P184-198.

[9] S. Holland, "Artificial Intelligence in Music Education: a critical review," *Contemp. Music Stud.*, vol. 20, 2013.

[10] M. Qiusi and H. A. Khattak, "Research on the Improvement Method of Music Education Level under the Background of AI Technology," *Mob. Inf. Syst.*, vol. 2022, no. 1, p. 7616619, Jan. 2022, doi: 10.1155/2022/7616619.

[11] M. Shang, "The Application of Artificial Intelligence in Music

Education," *Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)*, vol. 11644 LNCS, pp. 662–668, 2019, doi: 10.1007/978-3-030-26969-2_62.

[12] Q. Jiang, "[Retracted] Application of Artificial Intelligence Technology in Music Education Supported by Wireless Network," *Math. Probl. Eng.*, vol. 2022, no. 1, p. 2138059, Jan. 2022, doi: 10.1155/2022/2138059.

[13] I. A. Korsakova, "RISKS AND PROSPECTS FOR THE DEVELOPMENT OF DIGITAL TECHNOLOGIES AND ARTIFICIAL INTELLIGENCE IN MUSIC EDUCATION," *Muzykal'noe Iskusstv. i Obraz. (Musical Art Educ.*, pp. 24–34, 2024, Accessed: Mar. 10, 2025. [Online]. Available: http://musart-edu.ru/wp-content/uploads/2024/11/2024-2-24.pdf

[14] M. Della Ventura, "Exploring the Impact of Artificial Intelligence in Music Education to Enhance the Dyslexic Student's Skills," *Commun. Comput. Inf. Sci.*, vol. 1011, pp. 14–22, 2019, doi: 10.1007/978-3-030-20798-4_2.

[15] C. A. N. Mazlan *et al.*, "Discovery the intersection of performing arts in cultural tourism: a scoping review," *Discov. Sustain.*, vol. 6, no. 1, pp. 1–15, Dec. 2025, doi: 10.1007/S43621-025-00805-W/TABLES/5.

[16] R. Hidayatullah, R. Irawan, P. W. Alam, and C. A. N. Mazlan, "Jazz in Lampung: Community's Perspective, Social Practices, Cultural Performance, Organization, and Consequences," *Resital J. Seni Pertunjuk.*, vol. 25, no. 2, pp. 186–216, Aug. 2024, doi: 10.24821/RESITAL.V25I2.13664.

[17] C. A. N. Mazlan, M. H. Abdullah, N. S. N. Hashim, and N. A. Wahid, "Harmonizing heritage: A two-decade scoping bibliometric analysis of music in cultural tourism," *Multidiscip. Sci. J.*, vol. 7, no. 8, pp. 2025373–2025373, Feb. 2025, doi: 10.31893/MULTISCIENCE.2025373.

[18] D. S. H. A. Hassim and C. A. N. Mazlan, "Internal and External Factors that Influence Students' Persistent Participation in Music Learning," *Jurai Sembah*, vol. 4, no. 2, pp. 1–23, Dec. 2023, doi: 10.37134/JURAISEMBAH.VOL4.2.1.2023.

[19] R. Zakaria *et al.*, "Visualising and mapping a decade of literature on honey research: a bibliometric analysis from 2011 to 2020," *J. Apic. Res.*, vol. 60, no. 3, pp. 359–368, 2021, doi: 10.1080/00218839.2021.1898789.

[20] D. Moher, A. Liberati, J. Tetzlaff, and D. G. Altman, "Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement," *BMJ*, vol. 339, no. 7716, pp. 332–336, Jul. 2009, doi: 10.1136/BMJ.B2535.

[21] A. Pisali, N. S. N. Hashim, C. A. N. Mazlan, N. M. Nor, and I. M. Arsad, "Soft Clay Medium of Translating Children's Imaginary: Exploration of Stateless Children in Sabah, Malaysia," *Pakistan J. Life Soc. Sci.*, vol. 22, no. 1, pp. 1307–1324, 2024, doi: 10.57239/pjlss-2024-22.1.088.

[22] C. A. N. Mazlan *et al.*, "Technological Advancements in Bite-Sized Learning: Developing a Framework for Basic Jazz Guitar Reharmonization Techniques," *J. Adv. Res. Appl. Sci. Eng. Technol.*, vol. 43, no. 1, pp. 243–250, Jan. 2025, doi: 10.37934/ARASET.43.1.243250.

[23] Y. Yang, M. Lagisz, and S. Nakagawa, "Enriching meta-analyses through scoping review, bibliometrics, and alternative impact metrics: Visualizing study characteristics, hidden risk of bias, societal influence, and research translation," Dec. 2023, Accessed: Apr. 07, 2025. [Online]. Available: https://arxiv.org/abs/2312.01643v1

[24] S. Pirri, V. Lorenzoni, and G. Turchetti, "Scoping review and bibliometric analysis of Big Data applications for Medication adherence: An explorative methodological study to enhance consistency in literature," *BMC Health Serv. Res.*, vol. 20, no. 1, pp. 1–23, Jul. 2020, doi: 10.1186/S12913-020-05544-4/FIGURES/5.

[25] C. A. N. Mazlan, M. H. Abdullah, N. S. Nor Hashim, and N. Abdul Wahid, "Music in cultural tourism: insights from a dual approach of scoping review and bibliometric analysis," *Humanit. Soc. Sci. Commun. 2025 121*, vol. 12, no. 1, pp. 1–17, Apr. 2025, doi: 10.1057/s41599-025-04847-3.

[26] R. Cabrera, A. Carrion, and M. Huerta, "A Scoping Review and Bibliometric Analysis of BBC Micro:Bit Research," *EDUNINE 2024 - 8th IEEE World Eng. Educ. Conf. Empower. Eng. Educ. Break. Barriers through Res. Innov. Proc.*, 2024, doi: 10.1109/EDUNINE60625.2024.10500601.

[27] G. Parker, S. Hunter, K. Born, and F. A. Miller, "Mapping the Environmental Co-Benefits of Reducing Low-Value Care: A Scoping Review and Bibliometric Analysis," *Int. J. Environ. Res. Public Health*, vol. 21, no. 7, p. 818, Jul. 2024, doi: 10.3390/IJERPH21070818/S1.

[28] M. Ertz and S. Leblanc-Proulx, "Sustainability in the collaborative economy: A bibliometric analysis reveals emerging interest," *J. Clean. Prod.*, vol. 196, pp. 1073–1085, Sep. 2018, doi: 10.1016/J.JCLEPRO.2018.06.095.

[29] P. Mongeon and A. Paul-Hus, "The journal coverage of Web of Science and Scopus: a comparative analysis," *Scientometrics*, vol. 106, no. 1, pp. 213–228, Jan. 2016, doi: 10.1007/S11192-015-1765-5/METRICS.

[30] A. D. Moher D, Liberati A, Tetzlaff J, "PRISMA 2009 Flow Diagram," *The PRISMA statement*, vol. 6. p. 1000097, 2009.

[31] M. Pourkiaei and A. C. Romain, "Scoping review of indoor air quality indexes: Characterization and applications," *J. Build. Eng.*, vol. 75, p. 106703, Sep. 2023, doi: 10.1016/J.JOBE.2023.106703.

[32] C. A. N. Mazlan *et al.*, "Exploring the integration of bite-sized learning: A scoping review of research in education and related disciplines," *Contemp. Educ. Technol.*, vol. 15, no. 4, p. ep468, Oct. 2023, doi: 10.30935/CEDTECH/13622.

[33] M. R. Ramdan *et al.*, "The Impact of Applying the Servant Leadership Style among Educators in Educational Institutes: A Scoping Review," *TEM J.*, vol. 13, no. 2, pp. 1068–1079, 2024, doi: 34TUhttps://doi.org/10.18421/TEM132-21U34T.

[34] C. Verdejo, L. Tapia-Benavente, B. Schuller-Martínez, L. Vergara-Merino, M. Vargas-Peirano, and A. María Silva-Dreyer, "What you need to know about scoping reviews," *Medwave*, vol. 21, no. 2, 2021, doi: 10.5867/medwave.2021.02.8144.

[35] A. D. Samala, L. Bojic, D. Bekiroğlu, R. Watrianthos, and Y. Hendriyani, "Microlearning: Transforming Education with Bite-Sized Learning on the Go—Insights and Applications," *Int. J. Interact. Mob. Technol.*, vol. 17, no. 21, p. 4, Nov. 2023, doi: 10.3991/IJIM.V17I21.42951.

[36] A. Ahmi, "biblioMagika®," 2024. https://bibliomagika.com/ (accessed Aug. 13, 2024).

[37] S. Imam Koning, M. N. Nasrifan, N. W. Md. Junus, and N. Nor Azmi, "Integrasi aktiviti muzikal dalam pengajaran dan pembelajaran matematik: Satu kajian literatur," *J. Pendidik. Bitara UPSI*, vol. 14, pp. 92–102, Apr. 2021, doi: 10.37134/BITARA.VOL14.SP.10.2021.

[38] M. N. Nasrifan and Z. L. Saidon, "DEVELOPING AND TESTING A SELF-LEARNED INTERACTIVE MULTIMEDIA COURSEWARE FOR MUSIC AURAL LESSON," *IICACS Int. Interdiscip. Conf. Arts Creat. Stud.*, pp. 120–127, Aug. 2016, Accessed: Jul. 19, 2024. [Online]. Available: https://conference.isi-ska.ac.id/index.php/iicacs/article/view/86

[39] Z. Hong Yun, Y. Alshehri, N. Alnazzawi, I. Ullah, S. Noor, and N. Gohar, "A decision-support system for assessing the function of machine learning and artificial intelligence in music education for network games," *Soft Comput.*, vol. 26, no. 20, pp. 11063–11075, Oct. 2022, doi: 10.1007/S00500-022-07401-4/FIGURES/5.

[40] Y. Hu, "Striking a Chord with AI: A Case Study of Operational and Managerial Transformations in A Music Education Insinuation," Nov. 2023, doi: 10.4108/EAI.8-9-2023.2340186.

[41] J. Holster, "Augmenting Music Education through AI: Practical Applications of ChatGPT," *Music Educ. J.*, vol. 110, no. 4, pp. 36–42, Jun. 2024, doi: https://doi.org/10.1177/00274321241255938.

[42] J. Sang, "The Intersection of Technology and Art : A Study on AI-Driven CTCL Music Teaching Paradigm," no. 2, pp. 0–28, 2024.

[43] A. R. Safian *et al.*, "Design and Development of e-MARZ: A Practical Guide to Implementing Cost-effective VR Experiences for Music Education," *J. Adv. Res. Appl. Sci. Eng. Technol.*, vol. 40, no. 1, pp. 129–145, Feb. 2024, doi: 10.37934/ARASET.40.1.129145.